# The Field with One Element

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**Abstract**

$\mathbb{F}_1$ does not exist. But it should. If it did, $\operatorname{Spec}\mathbb{Z}$ would be a curve over it, the Frobenius would exist globally, and the Riemann Hypothesis would follow from the Weil conjectures. The field with one element is the name of the missing grammar.

## 1 The Idea

A field must have at least two elements: 0 and 1. The field axioms require additive and multiplicative inverses, and $0 \neq 1$. So $|\mathbb{F}| \geq 2$. There is no field with one element.

And yet.

Consider the analogy between $\mathbb{Z}$ and $\mathbb{F}_q[t]$. Both are principal ideal domains. Both have a theory of primes (integers vs irreducible polynomials). Both have zeta functions:

$$\zeta_{\mathbb{Z}}(s) = \prod_p \frac{1}{1-p^{-s}}, \qquad \zeta_{\mathbb{F}_q[t]}(s) = \prod_{f \text{ irred}} \frac{1}{1-|f|^{-s}}.$$

Over $\mathbb{F}_q$, the ring $\mathbb{F}_q[t]$ is the coordinate ring of the affine line $\mathbb{A}^1_{\mathbb{F}_q}$. The integers $\mathbb{Z}$ should be the coordinate ring of $\mathbb{A}^1_{\mathbb{F}_1}$: the affine line over the field with one element.

If $\mathbb{F}_1$ existed, $\operatorname{Spec}\mathbb{Z}$ would be a curve. Not just a scheme. A curve. With a smooth compactification: $\overline{\operatorname{Spec}\mathbb{Z}}$. And over a curve, the Weil machinery applies.

## 2 What $\mathbb{F}_1$ Would Give

Over a smooth projective curve $C/\mathbb{F}_q$, the Riemann Hypothesis holds: the eigenvalues of Frobenius on $H^1_{\text{ét}}(C, \mathbb{Q}_\ell)$ have absolute value $q^{1/2}$. This is proved.

The Riemann Hypothesis for $\zeta(s)$ is the Riemann Hypothesis for the curve $\overline{\operatorname{Spec}\mathbb{Z}}$ over $\mathbb{F}_1$. The zeros of $\zeta(s)$ are the eigenvalues of the $\mathbb{F}_1$-Frobenius on $H^1_{\mathbb{F}_1}(\overline{\operatorname{Spec}\mathbb{Z}}, \mathbb{Q})$. By the Weil bound: absolute value $1^{1/2} = 1$, corresponding to $\operatorname{Re}(s) = 1/2$.

This is not a proof. It is a translation. The proof would require:

1. A rigorous definition of $\mathbb{F}_1$ and geometry over it.
2. A construction of $H^1_{\mathbb{F}_1}$.
3. A proof that the Weil bound holds in this setting.

None of the three is complete. All three are active.

## 3 Approaches

**Smirnov (1992).** Noted the analogy between $\mathbb{Z}$ and $\mathbb{F}_q[t]$ and proposed that $\mathbb{F}_1$ should be the "field of characteristic 1." Did not define it.

**Manin (1995).** Proposed that $\operatorname{Spec}\mathbb{F}_1 = \{*\}$ is the absolute point, and that all algebraic geometry is relative to it. Introduced the slogan: finite sets are vector spaces over $\mathbb{F}_1$.

**Connes–Consani (2010–).** Defined geometry over $\mathbb{F}_1$ using $\Lambda$-rings and semirings. Connected $\mathbb{F}_1$ to the Bost–Connes system [1]: the symmetry group $\hat{\mathbb{Z}}^* \cong \operatorname{Gal}(\mathbb{F}_1^{\text{ab}}/\mathbb{F}_1)$ acts on the KMS states at $\beta > 1$. The base change from $\mathbb{F}_1$ to $\mathbb{Z}$ is the functor $\mathbb{Z} \otimes_{\mathbb{F}_1} (-)$.

**Toën–Vaqué (2009).** Defined $\mathbb{F}_1$ via monoids: an $\mathbb{F}_1$-algebra is a pointed monoid. A scheme over $\mathbb{F}_1$ is a sheaf of monoids. $\mathbb{Z} = \mathbb{F}_1[x, x^{-1}, \ldots]$ in this language.

**Borger (2009).** $\Lambda$-rings: $\mathbb{F}_1$-algebras are rings with a compatible family of Frobenius lifts $\psi^p$ for all primes $p$. The Adams operations give the $\mathbb{F}_1$-structure. $\operatorname{Spec}\mathbb{Z}$ carries a natural $\Lambda$-structure. This is the most developed approach.

# 4 The Curve

If $\operatorname{Spec}\mathbb{Z}$ is a curve over $\mathbb{F}_1$, its compactification must include the archimedean place.

The archimedean place is the point at infinity. This is Arakelov geometry [5]: compactifying $\operatorname{Spec}\mathbb{Z}$ by adjoining the archimedean place.

The curve $\overline{\operatorname{Spec}\mathbb{Z}}$ has:

- One point for each prime $p$: the closed point $p$.
- One point at infinity: the archimedean place.
- Genus: unknown. Conjectured to be 0 (like $\mathbb{P}^1$).

If the genus is 0, the Riemann Hypothesis would say the zeros of $\zeta$ have the right absolute value for a genus-0 curve, consistent with the known analytic results.

# 5 What the $\mathbb{F}_1$ Grammar Would Be

In the language of the grammar paper [2]:

The current grammar of arithmetic is the cohomological grammar of schemes over $\mathbb{Z}$. This grammar has local Frobenii ($\varphi_p$ at each prime) but no global Frobenius. It is locally complete and globally incomplete.

The $\mathbb{F}_1$ grammar would be the cohomological grammar of schemes over $\mathbb{F}_1$. In this grammar, $\operatorname{Spec}\mathbb{Z}$ is a curve, and the global Frobenius is the $\mathbb{F}_1$-Frobenius: a single canonical symmetry acting on the whole curve.

The local-to-global passage [3] would be solved by the new base: local Frobenii at each prime cohere globally because they are all pullbacks of the $\mathbb{F}_1$-Frobenius.

The wall would not be climbed. The base would be changed. From the new base, the wall does not appear.

*Remark* 1 (What does not yet exist). The $\mathbb{F}_1$ grammar is not yet a grammar. It is a sketch of one. The definition of $\mathbb{F}_1$ is not agreed upon. The cohomology theory $H^1_{\mathbb{F}_1}$ has not been constructed. The Weil bound has not been proved in any $\mathbb{F}_1$ setting. This paper describes what would be true if the grammar existed. It is not a proof. It is an anatomy of the absence.

# References


[1] Rincón, D. (2026). Bost–Connes: Zeta as Partition Function. Preprint.

[2] Rincón, D. (2026). Mathematics as Cohomological Grammar. Preprint.

[3] Rincón, D. (2026). One Wall: The Local-to-Global Conjecture. Preprint.

[4] Rincón, D. (2026). The Missing Frobenius. Preprint.

[5] Rincón, D. (2026). The Weil Conjectures: Proof of Concept for RH. Preprint.

[6] Rincón, D. (2026). Noncommutative Geometry and the Riemann Hypothesis. Preprint.